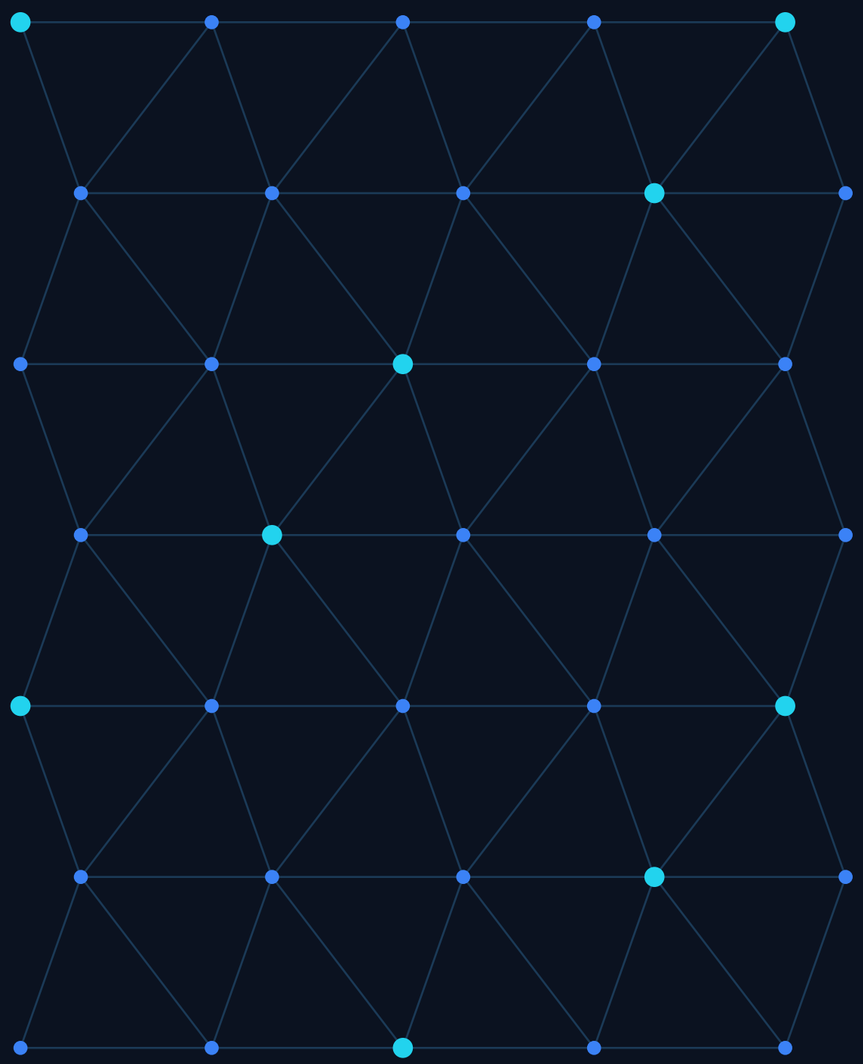
AGENTIC AI | DISASTER RESILIENCE

Synthetix-Grid

Autonomous Decoupled Multi-Agent Swarm for Block-Level Urban Climate Resilience

Track: Agentic AI Engineer



Presented by

Madhumitha.D

02 Problem Statement & Context

THE CHALLENGE

Micro-level urban disasters (flash floods, localized grid failures, hyper-local heat islands) escalate faster than human operators can triage data.

CURRENT LIMITATION

Centralized dashboards need manual triage, causing a 45-60 minute decision delay. Static LLMs lack live tool execution and dynamic task decomposition.

OUR IMPACT

Autonomous triage, route generation and emergency dispatch in under 60 seconds.

TRADITIONAL METHOD

Data Stream → **Human Manual Triage** → **Delayed Dispatch**

45-60 min

decision delay during the emergency window

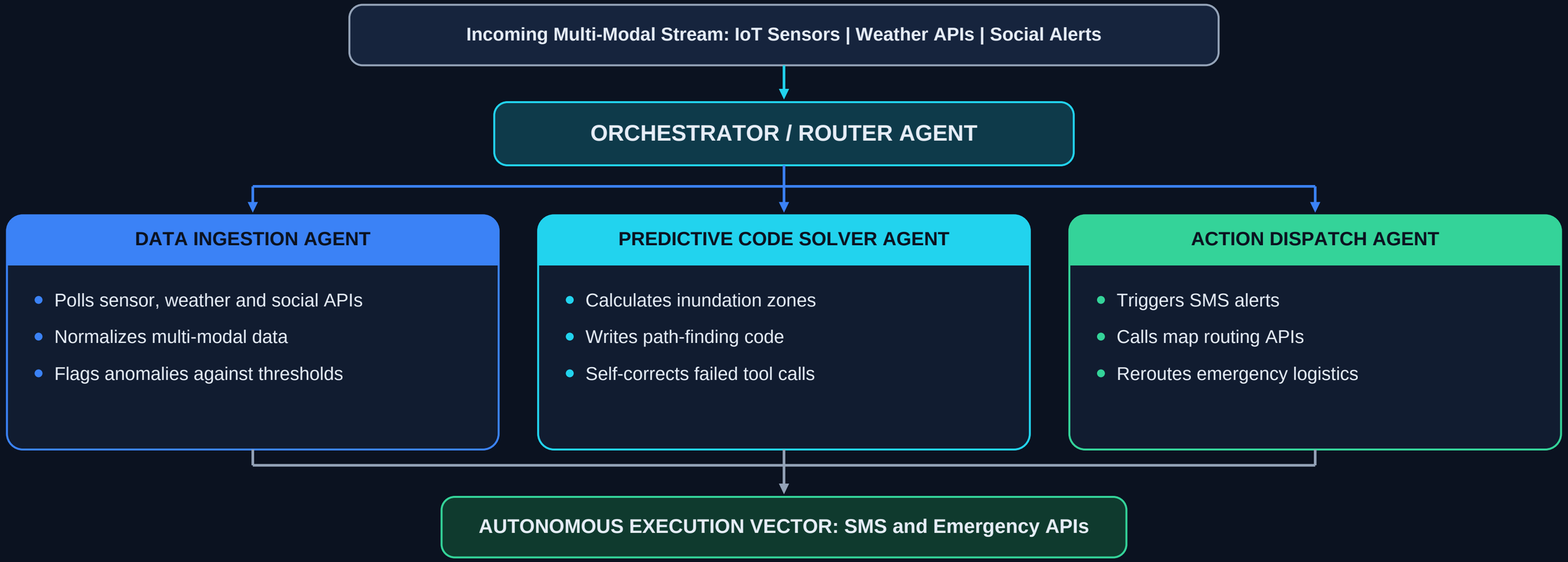
SYNTHETIX-GRID

Data Stream → **Autonomous Swarm** → **Instant Dispatch**

< 60 sec

autonomous end-to-end execution

03 Innovative Solution Architecture



CORE MECHANISMS

- **Continuous data polling**
- **Dynamic execution paths**
- **Self-correcting tool calls**

04 System Architecture & Technical Stack

LAYER 4 | PRESENTATION

Next.js 14 + TailwindCSS

Real-time agent trace dashboard for live monitoring of every agent action

LAYER 3 | ORCHESTRATION

LangGraph / AutoGen

Stateful execution graphs, supervisor routing and agent memory

LAYER 2 | INTELLIGENCE

Llama 3 70B (Groq) + Claude | ChromaDB

Sub-second sub-agent reasoning, Claude for planning, vector store for historical disaster logs

LAYER 1 | DATA INGESTION

FastAPI + Async Python Workers

Asynchronous microservices polling sensor, weather and social-signal APIs

Data flows upward: ingestion > intelligence > orchestration > dashboard

05 Core User Journey & Agent Trace Log

1 STEP 1: TRIGGER

Ingestion Agent

detects a rainfall sensor spike (> 80 mm/hr) and flags the anomaly.

2 STEP 2: REASONING

Predictive Solver Agent

pulls elevation maps, writes a path-finding script and calculates flood zones.

3 STEP 3: EXECUTION

Action Dispatch Agent

triggers SMS alerts and reroutes emergency logistics via map APIs.

LIVE AGENT TRACE (illustrative sample)

```
[00:00.0]   INGESTION      Sensor B-14 reads 86 mm/hr, above the 80 mm/hr threshold
[00:00.4]   ORCHESTRATOR   Anomaly flagged, task routed to Predictive Solver
[00:02.1]   SOLVER         Elevation maps loaded, inundation zones computed
[00:04.8]   SOLVER         Safe-route script generated for relief vehicles
[00:06.0]   DISPATCH       SMS alerts queued to local authorities
[00:07.2]   DISPATCH       Emergency vehicles rerouted via OpenStreetMap APIs
```

06 Feasibility, Scalability & Impact

98%

Target reduction in response latency

OPEN-SOURCE CORE

LangGraph + FastAPI + ChromaDB

EDGE-DEPLOYABLE

Hardware-agnostic core

OPERATIONAL FEASIBILITY

Built on open-source frameworks (LangGraph, FastAPI, ChromaDB) and accessible LLM endpoints.

COST & TOKEN OPTIMIZATION

Fallback routing to lightweight local models cuts API cost and avoids token exhaustion under high load.

SAFETY & RELIABILITY

Confidence thresholds with human override for high-impact actions; every decision is logged in the agent trace.

FUTURE SCOPE

Native edge deployment on Raspberry Pi / Jetson modules for offline, low-power operation.